AMITY UNIVERSITY KOLKATA

Weekly Progress Report

For the Week Commencing 26 October 2026

Enrollment No. A914138123021

NAME: UJAN SEN

Program: B.Sc Data Science Honors/Research

Company Name :
(Please mention if Applicable)

Faculty Guide's Name : DR. ANWESHA LAW

Industry Guide's Name : Not Applicable
(Please mention if Applicable)

Internship Mode : Offline

Project Title:
Route-Aware Machine Learning Framework for GPS-Based ETA Prediction

Targets for the week:
Complete work for: ETA features and route profiles. Built progress, speed, route-bin, and congestion features from the cleaned and snapped sessions.

Achievements for the week

Created route profiles with average speed, session support, coordinates, and congestion values for ETA prediction.

Future work plans

Continue the next pipeline stage, validate the outputs, and document decisions for the project report.

Week's Summary

Days / Time	
Monday	26 Oct: Calculated monotonic route progress. The feature table combines route position with observed movement conditions.
Tuesday	27 Oct: Derived speed from progress and timestamps. Speed is calculated from consecutive progress and timestamp values.
Wednesday	28 Oct: Assigned 120-metre route bins. The bins make sessions comparable at fixed route distances.
Thursday	29 Oct: Aggregated profile speeds and support counts. Support counts show how much historical data backs each profile value.
Friday	30 Oct: Inspected congestion and choke-point candidates. The resulting profiles provide the inputs needed by the ETA models.